

Student Information System

Individual Presentation Handout

UC-06 — Generate Report

Requirement · Use Case · Activity Diagram · Sequence Diagram · Communication Diagram
April 2025

1. Requirement — BR-SIS-06

This requirement defines the report generation capability of the SIS. It is Priority 3 because it depends on the completion of enrollment and grading workflows (UC-01, UC-05). The requirement is fully traceable to Use Case UC-06, Activity Diagram AC-06, Sequence Diagram SC-06, and Communication Diagram CC-06.

Field	Detail
Req ID	BR-SIS-06
Priority	3 — Should Have (administrative reporting)
Module	Reporting & Analytics
Description	The system shall allow an authenticated Admin to generate an Academic Summary Report for a selected student, covering enrollment history, grades, and GPA. The system shall compile data from the Student, Enrollment, Grade, and Course tables and produce a formatted report delivered to the Admin.
Actor	Admin (primary)
Pre-condition	Admin is logged in. At least one Student record exists in the system. Enrollment and Grade records exist for the selected student.
Post-condition	A Report record is created (AdminID, StudentID, ReportType, GeneratedDate). The formatted report is displayed to the Admin. Optionally, a Notification record is created for the student.
Source	Academic Registrar / Functional Requirements Specification v1.0
Date Reviewed	April 2025
Approved By	Academic Registrar Office

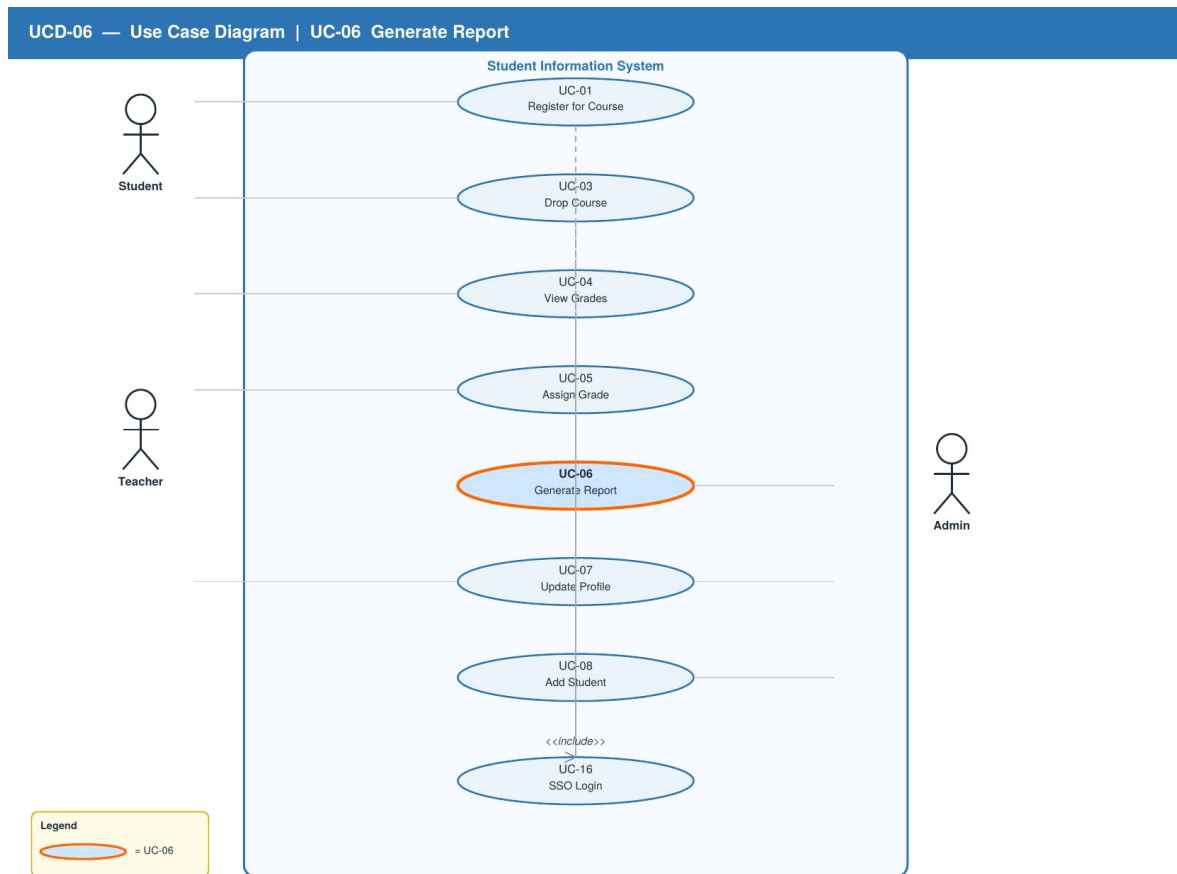
2. Use Case Table — UC-06

The use case below expands BR-SIS-06 into a structured, testable specification. It defines all actors, triggers, flows, and exceptions. The Main Flow maps step-by-step to the database operations shown in the sequence diagram (SC-06).

Field	Detail
Use Case ID	UC-06
Use Case Name	Generate Report
Requirement Ref	BR-SIS-06
Actor(s)	Admin (primary) System (secondary — compiles and formats data)
Trigger	Admin selects a student and clicks "Generate Report".
Pre-conditions	1. Admin is authenticated via SSO. 2. Target student exists in the Student table. 3. At least one Enrollment and Grade record exists for the student.
Post-conditions	1. Report record inserted: (ReportID, AdminID, StudentID, ReportType="Academic Summary", GeneratedDate). 2. Report displayed to Admin with enrollment list, grades, and computed GPA. 3. Optional Notification record inserted for the student.
Main Flow	1. Admin opens the Reports section. 2. System verifies session token. 3. Admin searches for a student by name or ID. 4. System queries Student table and returns matches. 5. Admin selects a StudentID and chooses report type (Academic Summary). 6. System queries Enrollment table for all enrollments by StudentID. 7. System queries Grade table for all grades linked to those enrollments. 8. System queries Course table to retrieve CourseName and Credits. 9. System computes GPA from Grade values and credit weights. 10. System inserts a Report record (AdminID, StudentID, ReportType, GeneratedDate). 11. System renders and displays the formatted Academic Summary report. 12. Admin views and optionally exports the report.
Alternative Flow	A1 — No records found: Step 6 returns empty set — system displays "No enrollment data available for this student". A2 — Partial data: Grades missing for some courses — GPA computed from available data; missing entries shown as N/A. A3 — Session expired: Step 2 fails — redirect to login.
Exception Flow	E1 — Database error at Step 10: rollback Report insert, display "Report generation failed — please retry". E2 — Network timeout: display error, no record committed.
Includes	UC-16 (Authenticate / SSO Login)
Extended By	UC-07 (Update Profile) UC-04 (View Grades)
Priority	3 — Should Have

3. Use Case Diagram — UCD-06 (UC-06 in context)

UC-06 (Generate Report) is shown in context with the other use cases. The diagram shows the «include» dependency on UC-16 (SSO Login) — every report generation action requires prior authentication. The Student is an indirect actor whose data is consumed by the report.



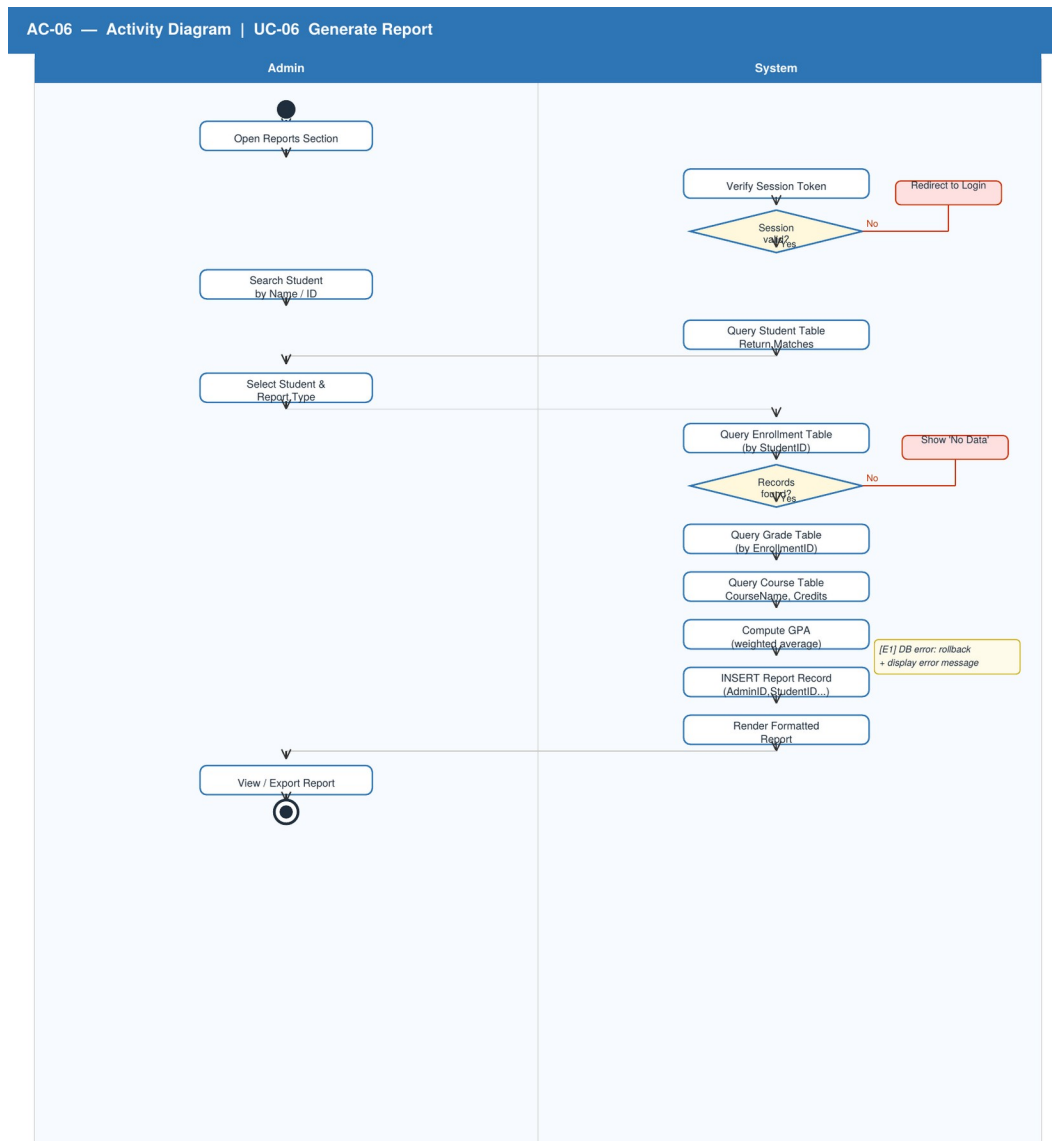
UCD-06 — Use Case Diagram (UC-06 highlighted in context)

4. Activity Diagram — AC-06

The activity diagram shows the complete flow for UC-06 including all decision points, alternate paths, and exception handling. Key decision gates are: (a) Is the session valid? and (b) Are enrollment records found? Both lead to distinct YES/NO branches before the Report record is created.

ERD attributes exercised:

- Report: ReportID, AdminID (FK), StudentID (FK), ReportType, GeneratedDate
- Enrollment: EnrollmentID, StudentID (FK), CourseID (FK), Status
- Grade: GradeID, EnrollmentID (FK), GradeValue, GradeDate
- Course: CourseID, CourseName, Credits, Semester
- Notification: StudentID (FK), NotificationText, NotificationDate



AC-06 — Activity Diagram — Generate Report (UC-06)

5. Sequence Diagram — SC-06

The sequence diagram shows the chronological message flow between 9 components. Each numbered message corresponds to an exact step in the UC-06 Main Flow. Database messages show the actual SQL column names from the ERD to make the data model explicit.

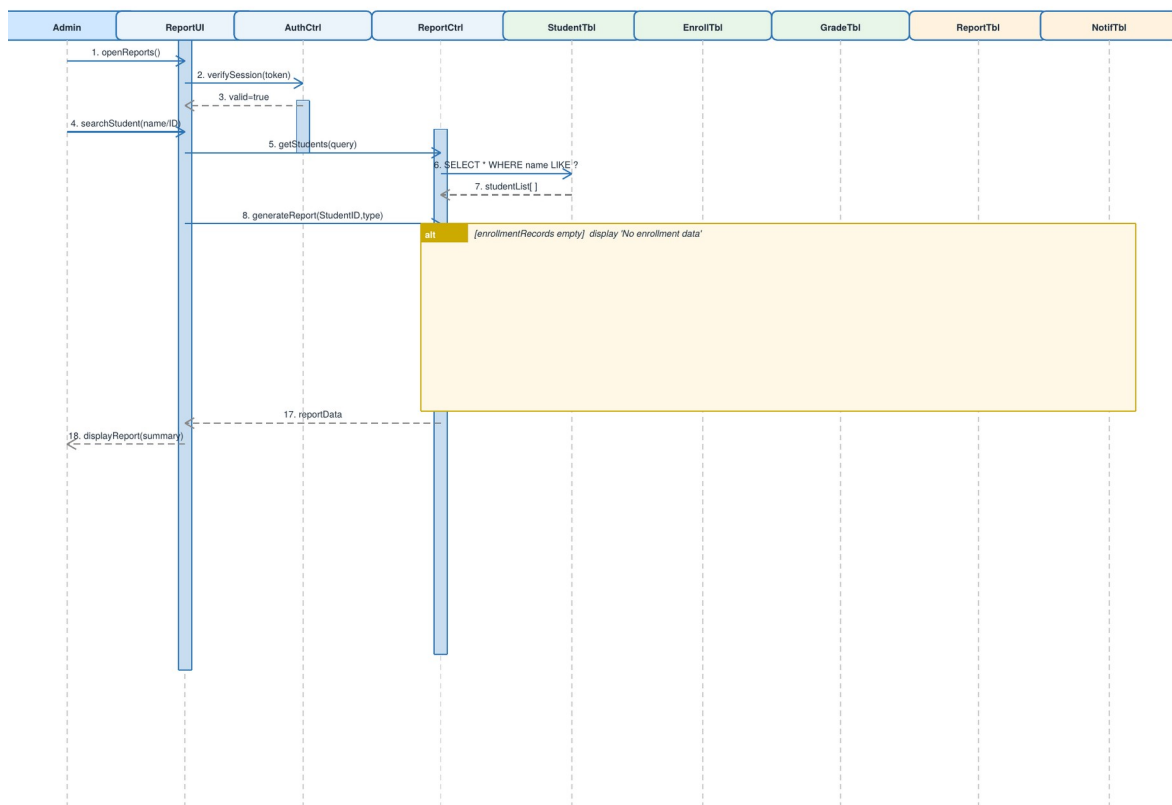
Component responsibilities:

- Auth Controller — validates session token against UserAccount table (UserID, Role)
- Report Controller — orchestrates all business logic: student lookup, data aggregation, GPA computation, record creation
- Student Table — source of StudentID, FullName, Email used in report header
- Enrollment Table — queried by StudentID to retrieve all CourseID + Status pairs

- Grade Table — queried by EnrollmentID to retrieve GradeValue and GradeDate
- Course Table — queried by CourseID to retrieve CourseName and Credits for GPA weighting
- Report Table — receives INSERT (AdminID FK, StudentID FK, ReportType, GeneratedDate)
- Notification Table — receives INSERT (StudentID FK, NotificationText, NotificationDate)

The [ALT] fragment covers the "no enrollment records found" guard condition.

SC-06 — Sequence Diagram | UC-06 Generate Report



SC-06 — Sequence Diagram — Generate Report (UC-06)

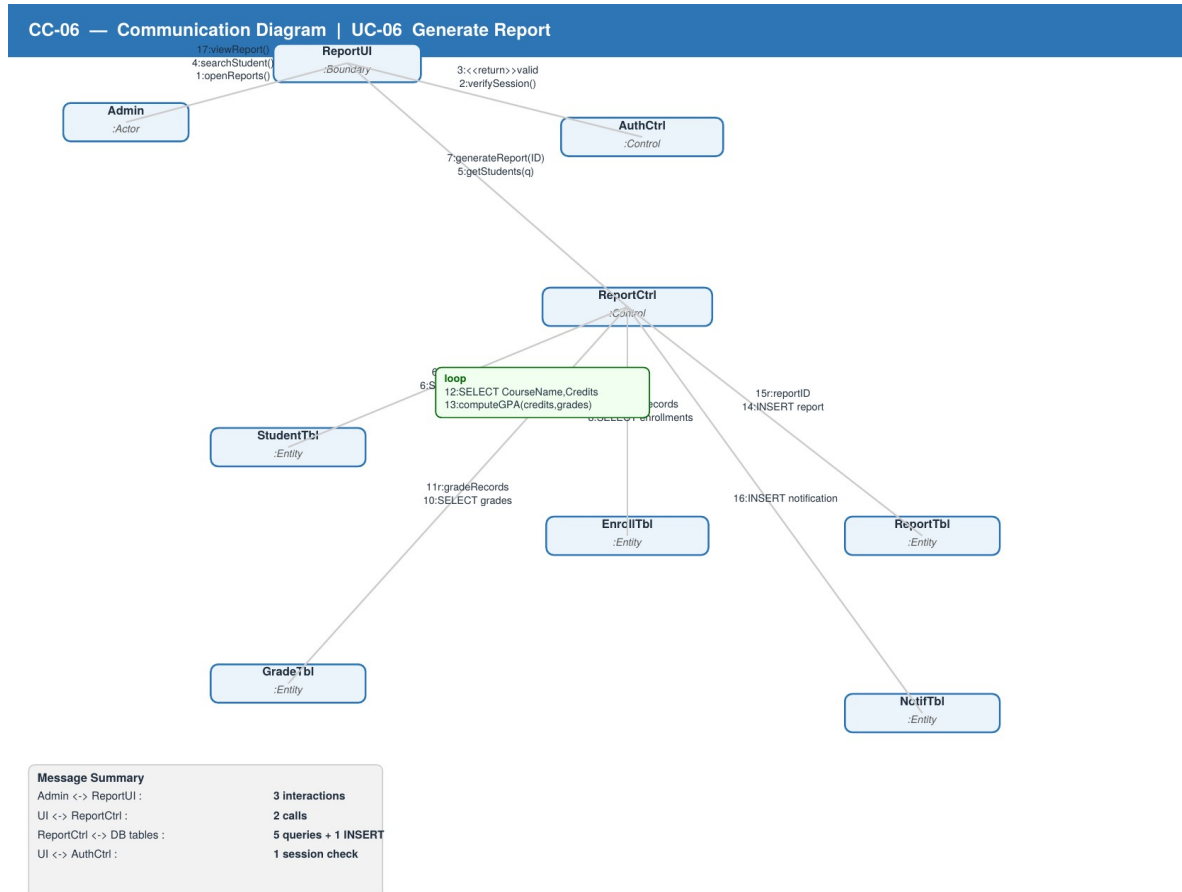
6. Communication Diagram — CC-06

The communication diagram (derived from SC-06) re-presents the same interactions as structural object relationships rather than a timeline. Numbered messages (1–17) match exactly the sequence in SC-06. The diagram emphasises which objects collaborate directly and how many messages flow across each link.

Key object relationships visible in CC-06:

- Admin ↔ ReportUI — 3 direct interactions (open, search, view report)
- UI ↔ ReportController — 2 requests (getStudents, generateReport)
- ReportController ↔ StudentTable — 1 SELECT query (student lookup)

- ReportController ↔ EnrollmentTable — 1 SELECT (all enrollments for StudentID)
- ReportController ↔ GradeTable — 1 SELECT (grades for each EnrollmentID)
- ReportController ↔ ReportTable — 1 INSERT (persist report metadata)
- ReportController ↔ NotificationTable — 1 INSERT (confirmation notification)
- UI ↔ AuthController — session verification (links to UC-16 «include»)



CC-06 — Communication Diagram — Generate Report (UC-06)

7. Traceability Summary

All artefacts in this handout are fully cross-referenced to the ERD and Requirements Specification.

Artefact	Maps To
BR-SIS-06 (Requirement)	Priority 3 · Functional Requirements Specification v1.0
UC-06 (Use Case Table)	BR-SIS-06 · Includes UC-16 · Extended by UC-04, UC-07
UCD-06 (Use Case Diagram)	UC-06 shown in system context with all 3 actors (Admin, Student, Teacher)

AC-06 (Activity Diagram)	UC-06 main flow + 2 decision branches + exception paths
SC-06 (Sequence Diagram)	AC-06 steps 1-12 · ERD: Admin, Student, Enrollment, Grade, Course, Report
CC-06 (Communication Diagram)	SC-06 messages 1-17 · same ERD entities as SC-06
ERD Entities Used	Admin · Student · Enrollment · Grade · Course · Report · Notification
ERD Relationships Used	Admin--Generates->Report · Report--References->Student · Student--HasEnrollment->Enrollment · Enrollment--HasGrade->Grade · Course--IsIn->Enrollment